

Population organization and closed-loop attractors for obstacle avoidance under perturbation

Summary (226 words; 300-word maximum)

Motor systems must adjust ongoing actions when unexpected forces occur, yet continue toward a goal and respect physical constraints. How can a learned controller organize such corrections when obstacles narrow the available route? We asked whether reward-based training alone can produce structured obstacle avoidance and whether the resulting activity offers an organizational comparison with biological motor control. We trained a proximal policy optimization (PPO) controller to move a point mass through a corridor with two obstacles. Its feedforward multilayer perceptron (MLP) actor had two 256-unit hidden layers. We analyzed the activity of a 256-unit hidden-unit population using principal component analysis (PCA), clustering, kinematic tuning, linear decoding, and closed-loop dynamical analysis of the policy coupled to the physics. Moderate lateral perturbations produced path changes under 0.1 m while the controller continued toward the goal. The first two principal components explained 71% of activity variance; activity followed phase-ordered routes shared across perturbation conditions and separated by detour. At the obstacle band, the closed loop had two stable lateral fixed points, one per route ($x^*=3.32$ m and 6.71 m), with similar relaxation times (1.42 s and 1.34 s) and clearances (0.43 m and 0.46 m). These results show that reward learning can yield structured, locally stabilizing routes in this model. The resemblance to biological reaching is organizational, not evidence of neural equivalence; results are from one training seed.

Additional detail: question and approach

Success at reaching a target does not reveal how feedback corrects movement after an unexpected force. The correction must preserve progress toward the goal while respecting the obstacle boundary. We therefore examined route structure and population activity, not only task success. ?Population? here means the 256 artificial units in one hidden layer?not neurons recorded from an animal. The feedforward actor processes each observation independently and has no recurrent memory. Accordingly, the fixed points are not equilibria of recurrent hidden activity [2]; they are fixed points of the complete policy?environment closed loop.

Model and perturbation protocol

A proximal policy optimization actor?critic was trained for 3 million environment steps. The actor was a feedforward multilayer perceptron with two 256-unit hidden layers and selected an action from the current state at each timestep. We evaluated a baseline and six perturbation conditions: lateral force impulses in three magnitudes to the left or right, each lasting 14 timesteps. We analyzed activity from a 256-unit hidden layer using principal component analysis, clustering, kinematic tuning, and linear decoding. These analyses characterize activity by movement phase and route and test how much movement information can be read from the population.

Additional detail: closed-loop results and biological scope

The first two principal components explained 40% and 31% of activity variance. At $y = 5$ m, the obstacle band, each route had one zero crossing in commanded lateral acceleration with a negative slope, consistent with local restoring behavior. Linearization gave real eigenvalues of magnitude below 1 for both routes, indicating stable, nonoscillatory fixed points at $x^* = 3.32$ m and 6.71 m. The points were within 0.11 m of observed completed-episode crossings. The two obstacles were centered at $x = 4.25$ m and $x = 5.75$ m, each with a 0.5 m radius; route clearances were 0.43 m and 0.46 m. Relaxation times (1.42 s and 1.34 s) describe local return in the model, not human response latency. Biological reaching also shows graded corrections to unexpected forces while movement continues, supported by coordinated muscle activity [1, 3, 4]. This is an organizational comparison, not evidence of neural or muscular equivalence. Results are from one training seed; fixed points vary across checkpoints. Multi-seed training and planned wrist-kinematic and six-muscle surface electromyography (EMG) comparisons in healthy adults and stroke survivors will test robustness and biological scope.

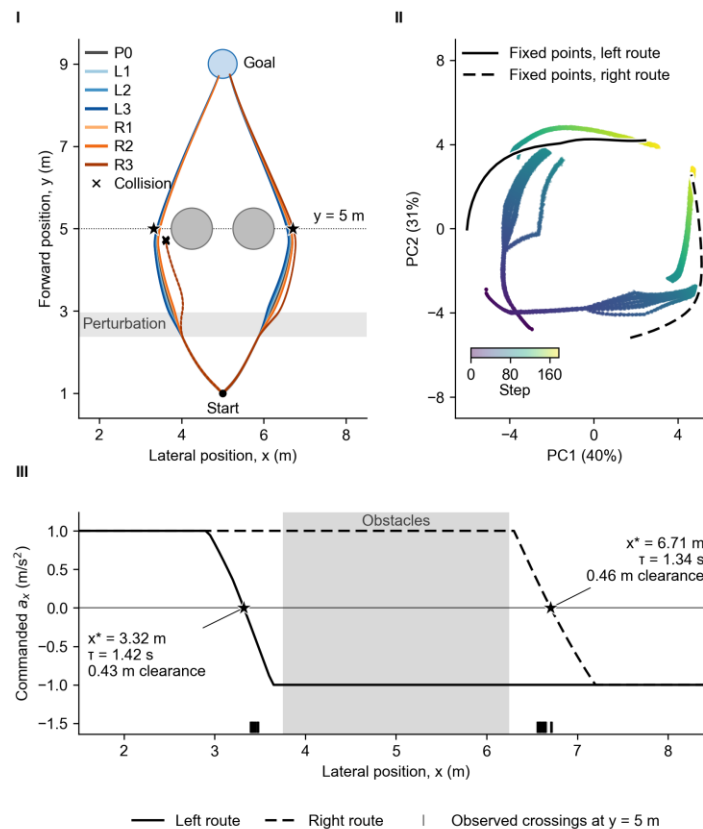


Figure 1. Obstacle avoidance and its closed-loop organization. (I) Baseline and six perturbation trajectories; crosses indicate collisions, observed only for the strongest rightward impulse on the left route; stars mark fixed points at $y = 5$ m. (II) Hidden-layer activity projected onto PC1/PC2 and colored by step, with fixed-point routes. (III) Commanded lateral acceleration at $y = 5$ m; each route has a zero crossing with negative slope. Ticks mark completed-episode crossings of the obstacle band.

References

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